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Which of a thousand tracks should stand in for my cell type? Label-efficient Bayesian discrimination of genomic foundation-model outputs

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Abstract

Motivation: Sequence-to-function foundation models such as AlphaGenome predict the effect of any variant on thousands of functional tracks, but a laboratory studying a particular cell type must still decide which track to trust. Matching tracks by cell-type name is unreliable, and many cell types are not profiled at all. A few variants can be measured, for example with a reporter assay, but it is unclear which variants to measure and how to turn a handful of measurements into a choice among about a thousand highly correlated tracks.
Results: Each track is a hypothesis that the measured effect is an unknown affine function of its prediction. We integrate out the calibration, form a posterior over tracks, keep metadata-matched tracks as the prior default, and measure next the variant on which the tracks disagree most. In two pre-registered tests on saturation mutagenesis reporter assays, eight chosen measurements beat eight random ones by 0.084 in Spearman correlation on 11 held-out Kircher et al. regions with the cell type hidden ($P = 0.0024$). On 125 new Siraj et al. loci, all four pre-registered hypotheses held. Where matched tracks existed, eight measurements did not harm them (mean +0.010; 3/125 loci worse by more than 0.05). Without them, eight measurements improved on the all-track mean by 0.048 ($P = 8 \times 10^{-13}$). With only related-cell tracks, sixteen improved on those tracks' mean by 0.041. With random measurements, the same inference fell below the label-free mean. Re-fitting the matched tracks to the few labels lowered accuracy at all 125 loci. All four hypotheses still held when each locus was scored by a model checkpoint that had not been trained on its region.
Availability and implementation: Code, pre-registrations and derived results are at `https://github.com/ydvislab/AuraGenome-V` under the MIT licence. [TODO: Zenodo archive DOI of the submitted version, package name]
Contact: [TODO: email]
Supplementary information: Supplementary data are available at *Bioinformatics* online.

Key words: variant effect prediction, foundation models, active model selection, Bayesian experimental design, massively parallel reporter assays

1. Introduction

Deep sequence-to-function models predict the regulatory consequences of non-coding variants from DNA sequence alone [Zhou and Troyanskaya, 2015, Avsec et al., 2021, Linder et al., 2025]. The most recent of them, AlphaGenome, outputs thousands of genome tracks for hundreds of human cell types and tissues, and scores any single-nucleotide variant on all of them [Avsec et al., 2026]. These predictions are accurate in aggregate. On the Kircher et al. saturation mutagenesis data [Kircher et al., 2019], used in the CAGI5 challenge [Shigaki et al., 2019], AlphaGenome is among the best available predictors. Accuracy at a single locus is less certain, and performance on personal expression variation is known to be weak [Sasse et al., 2023, Huang et al., 2023].

A practical question sits between the model and the bench: *which output track should a laboratory use for its cell type?* The usual answer is to pick tracks whose biosample name matches the cell line in which the variants will be studied. We found this unreliable.

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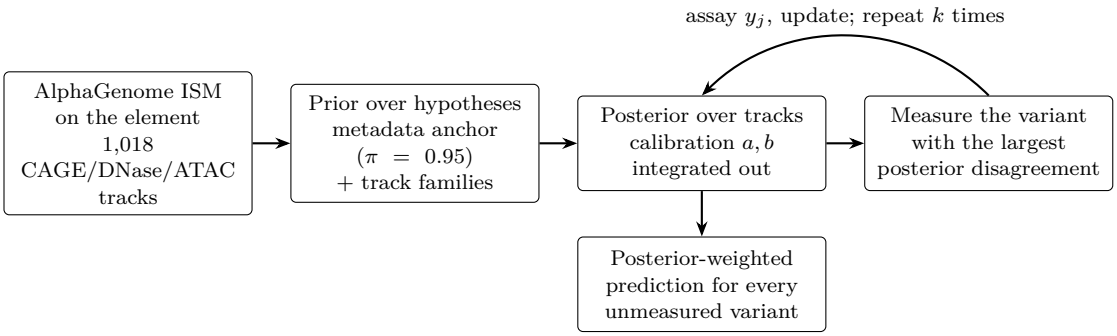


Fig. 1. Overview. Each AlphaGenome track is a hypothesis that the measured effect is an affine function of the track's predicted effect. With k measured variants, the method updates a posterior over hypotheses whose prior keeps the metadata-matched tracks as the default. It picks each next variant where the hypotheses disagree most and reports a posterior-weighted prediction.

Across the 1,018 CAGE, DNase and ATAC tracks AlphaGenome predicts, the name-matched track ranked between first and 27th by agreement with the measured effects in our development elements. Choosing a different track changed the conclusions. At the *PKLR* promoter in K562 cells, replacing K562 tracks by HepG2 tracks lowered the directional accuracy on significant variants from 0.96 to 0.69–0.85, and the two sets of tracks disagreed in sign on 13–32% of significant variants. Many cell types used in the laboratory, including patient-derived lines, are not profiled at all, and then there is no name to match.

Laboratories can, however, measure a few variants. Reporter assays, base editing or targeted allelic assays can cheaply test a panel of perhaps 8–16 variants before a larger study. This turns proxy choice into a label-efficient model selection problem: choose which variants to measure so that the measurements identify the track, or the mixture of tracks, that best predicts the rest. Two features make it different from standard active model selection [Karimi et al., 2021, Okanovic et al., 2025, Kay et al., 2025]. First, the candidates are regression outputs on arbitrary scales, so each track must be calibrated to the assay with an unknown slope and offset that a handful of labels cannot pin down. Second, the candidates are a thousand strongly correlated heads of one network, not a set of independently trained models.

We make three contributions. (i) A Bayesian formulation in which each track is a hypothesis with an affine calibration that is integrated out in closed form. The result is a posterior over tracks and a posterior-weighted prediction for unmeasured variants. A metadata anchor prior makes the method safe when matched tracks exist. (ii) A sequential design that measures the variant on which the posterior-weighted tracks disagree most, in the tradition of model discrimination [Box and Hill, 1967], applied here to about a thousand candidates. (iii) Two pre-registered evaluations on independent saturation mutagenesis data (Kircher et al., 11 held-out regions; Siraj et al., 125 new loci), plus a check of the dependence on the model's training exposure. These show when a few measurements help, when they do not, and that the obvious alternative, re-fitting the matched tracks to the same few labels, is harmful.

2. Methods

2.1. Problem

A regulatory element is studied in cell type c. Saturation mutagenesis gives a set V of single-nucleotide variants with measured effects y_j; we pretend that only k of them may be measured. For every variant, the foundation model gives a predicted effect $x_{t,j}$ on each of T tracks (here $T = 1{,}018$). We want a prediction $\hat{y}_j$ for every unmeasured variant, and the measurement panel $S \subset V$, $|S| = k$, that makes these predictions as accurate as possible. Accuracy is the Spearman correlation between $\hat{y}$ and y on variants not available for measurement.

2.2. Tracks as calibrated hypotheses

For each track we use the standardised prediction $z_{t,j} = x_{t,j}/\mathrm{sd}_j(x_{t,\cdot})$. It is scaled over the element but not centred, so the sign keeps the direction of the alternative allele. Hypothesis t states

$$\begin{aligned} &y_j = a + b\, z_{t,j} + \varepsilon_j, \\ &a \sim N(0, \alpha s^2), \quad b \sim N(0, \beta s^2), \quad \varepsilon_j \sim N(0, (1-\beta)s^2), \end{aligned} \tag{1}$$

where s^2 is the mean of y_S^2. Integrating out a and b gives the marginal likelihood of the measured panel,

$$y_S \mid t \sim N\Big(0,\ s^2\left[(1-\beta)I + \alpha \mathbf{1}\mathbf{1}^\top + \beta z_{t,S} z_{t,S}^\top\right]\Big), \tag{2}$$

and the posterior $p(t \mid y_S) \propto p(t)\, p(y_S \mid t)$. The predictive mean of an unmeasured variant under t, $m_{t,j} = E[y_j \mid t, y_S]$, follows from the Gaussian conditional. The reported prediction is the Bayesian model average $\hat{y}_j = \sum_t p(t \mid y_S)\, m_{t,j}$ [Hoeting et al., 1999]. We fixed $\alpha = 0.1$ and $\beta = 0.5$ before any evaluation and did not tune them.

2.3. Metadata anchor and track families

When the model profiles the element's cell type, the mean of the metadata-matched tracks is a strong default. We add it as one extra hypothesis, the *anchor*, holding prior mass π. Its slope prior is centred on a positive value, $b \sim N(\sqrt{\beta}\, s,\ (\beta/4)\, s^2)$, so that a negative slope has prior probability of about 2%. Single tracks share the remaining $1 - \pi$. To prevent near-duplicate tracks from dominating the prior, we cluster tracks by the correlation of their predictions (average linkage, $r \geq 0.9$; no labels used). The prior is split equally between clusters and then within them, so duplicating a track changes nothing. When no matched tracks exist, there is no anchor, and the method reduces to the track posterior with the family prior. The case $\pi = 0$ with a uniform prior is the version we pre-registered first (v1). The anchor version (v2, $\pi = 0.95$) was chosen on development elements only, frozen, and then tested (Section 2.7).

2.4. Choosing which variants to measure

The panel is built sequentially from a pool of variants that could be measured. Given the current posterior, the next variant maximises

the posterior-weighted variance of the hypotheses' predictive means, $\mathrm{Var}_{t\sim p(t|y_S)}[m_{t,j}]$, divided by the expected residual variance of that variant. This is the classical criterion for discriminating between models [Box and Hill, 1967], a member of the family of Bayesian experimental designs [Rainforth et al., 2024]. We call this panel *disc*. All panels are scored with the same inference, so panels differ only in which variants are measured.

2.5. Baselines

Other panels. We compare disc with random variants; the variants with the largest predicted effect under the all-track mean, a common practical choice; and the variants with the largest unweighted disagreement across tracks. *Label-free references.* These are the mean of all tracks; the mean of the metadata-matched tracks; and, when those are hidden, the mean of related-cell tracks. The best single track, chosen with all labels, is reported as a reference, not an upper bound. *Few-label regression.* Ridge regression and LASSO [Tibshirani, 1996] over all tracks, with the penalty chosen by leave-one-out on the same k labels. This is the protocol used for AlphaGenome on this benchmark, which there uses the full challenge training set. *Same-budget metadata controls.* We include `matched_cal`, an affine re-fit of the matched mean with the k labels, and `meta_ridge`, which calibrates the matched mean with a positive-slope prior and then fits ridge regression on the residuals over all tracks.

2.6. Data

Kircher et al. [Kircher et al., 2019] measured every possible substitution in 20 disease-associated regulatory elements by saturation mutagenesis MPRA. We used the 22 experiments measured in human cells (27,789 variants, 561–1,800 per experiment). Five experiments from five regions (*PKLR*, *F9*, *HNF4A*, *IRF6*, *BCL11A*) formed the development set. The other 17 experiments from 11 regions formed the test set; several of them were measured in cell lines AlphaGenome does not profile (for example SK-MEL-28 and glioblastoma lines). **Siraj et al.** [Siraj et al., 2026, 2025] measured all substitutions in 200-bp oligonucleotides around fine-mapped trait variants, in K562 and HepG2 cells. Of the 128 published elements, we used the reference background of 125. One was excluded because its reference background contains a deletion, and two because the oligo had no unique exact match in GRCh38. This left 74,975 substitutions, each measured in both cell types; the effect is the reported $\log_2$ allelic skew. **Predictions.** We scored every variant by in-silico mutagenesis with AlphaGenome (API, 1-Mb context) on all CAGE, DNase and ATAC tracks for human biosamples [Avsec et al., 2026]. These assays derive from ENCODE and FANTOM5 [ENCODE Project Consortium et al., 2020, The FANTOM Consortium and the RIKEN PMI and CLST (DGT), 2014]. The main analyses use the default distilled checkpoint (ALL_FOLDS).

Training data, test sets and homology. The method trains no model. AlphaGenome is used frozen, and the only labels are the k measured variants of the element being predicted. No measured effect from any other element, and no evaluation variant, enters the inference. Performance is reported on two independent test sets defined before analysis: 11 Kircher regions not used during development, and 125 Siraj loci not used at any stage of method design. No test region overlaps a development region; the nearest pair, one Siraj locus within 500 kb of the *IRF6* development element, shares only long-range model context. Two kinds of exposure remain. MPRA labels were never part of AlphaGenome's training, but its training tracks cover these genomic regions. We test the dependence on this exposure with checkpoints that held out the genomic fold of each locus (Section 3.4). The folds are the model developers' split of the genome into sets of intervals [TODO: state from the AlphaGenome/Borzoi split description how homologous sequence is assigned to folds; not yet verified]. Leave-one-out is not used for evaluation. It appears only inside the ridge and LASSO baselines, to choose a penalty from the k labels, and in a leave-one-element-out sensitivity check of π on the development set.

2.7. Evaluation protocol and pre-registration

In each of 200 repeats (seed 0), half of an element's variants form the pool that could be measured. Panels are built from the pool, and every method and reference is scored by Spearman correlation on the same variants outside the pool. Experiments that share a genomic region are averaged, and the analysis unit is the region (Kircher) or the locus (Siraj; its two cell types are averaged). We simulate three situations by hiding tracks. In C1 all tracks are kept. In C2 only the element's own cell line is hidden, and related-cell tracks remain. In C3 both are hidden, which approximates a cell type the model does not profile. Related cell types were listed from track metadata before any result was seen. Two pre-registrations were committed together with frozen code before the test data were analysed. *Kircher test (v1):* H1, disc beats random panels (one-sided Wilcoxon, $P < 0.05$); H2, disc beats the all-track mean by at least 0.03 in at least 7/11 regions; H3 (secondary), disc beats high-predicted-effect panels. *Siraj test (frozen v2):* P1, non-inferiority to the matched mean in C1 at $k = 8$ (one-sided 95% bootstrap lower bound above -0.02); P2, at most 15% of loci worse by more than 0.05 (one-sided 95% Clopper–Pearson upper bound); P3, improvement over the all-track mean in C3 at $k = 8$ of at least 0.03 with $P < 0.05$; P4, improvement over the related-cell mean in C2 at $k = 16$ with $P < 0.05$. The safety-net claim required P1, P2 and either P3 or P4. Analyses added after the pre-registrations are labelled as such.

3. Results

3.1. A few chosen measurements recover the proxy when the cell type is hidden

On the Kircher test regions with the element's own and related cell types hidden (C3), eight disc measurements gave a mean Spearman correlation of 0.370, against 0.286 for eight random measurements and 0.296 for the label-free all-track mean (Table 1, Fig. 2). The pre-registered H1 held: disc beat random panels by 0.084, 9/11 regions, $P = 0.0024$. H2 also held: disc beat the all-track mean by 0.074, 8/11 regions. H3 did not: the margin over high-predicted-effect panels was 0.031, $P = 0.18$. The gains were concentrated in elements with signal and room above the average. They included *HBG1*, where the posterior-weighted prediction exceeded the best single track, and *IRF4* in SK-MEL-28, a cell line the model does not profile. Elements with very few significant variants (*FOXE1*, 5; one *MYC* enhancer, 11) or with an all-track mean already near the best track (*TERT*) showed no gain.

With the same eight labels, regression over all tracks was much worse (post hoc). Disc beat random panels with LASSO by 0.110–0.182, depending on how all-zero LASSO fits are treated ($P \leq 0.0024$), and random panels with ridge by 0.116. On the disc panel itself, the Bayesian track posterior still beat LASSO by 0.034–0.048 and ridge by 0.023 ($P = 0.042$). So both the panel and the inference contribute.

Table 1. Kircher test set, C3 (own and related cell types hidden), $k = 8$, 11 regions. Spearman correlation on the common evaluation set. †Added after pre-registration (first 50 repeats); for LASSO the range covers three rules for all-zero fits.

Panel	Prediction	Mean ρ	Ours − this row	Regions > 0, one-sided P
disc	track posterior (ours)	0.370	—	—
random	track posterior	0.286	+0.084	9/11, 0.0024
high predicted effect	track posterior	0.338	+0.031	–, 0.18
static disagreement	track posterior	0.361	+0.009	–, –
none	all-track mean	0.296	+0.074	8/11, –
random	LASSO†	–	+0.110 to +0.182	9–10/11, ≤0.0024
random	ridge†	–	+0.116	8/11, 0.0068
disc	LASSO†	–	+0.034 to +0.048	8/11, ≤0.016
disc	ridge†	–	+0.023	8/11, 0.042
all labels	best single track (reference)	0.454	—	—

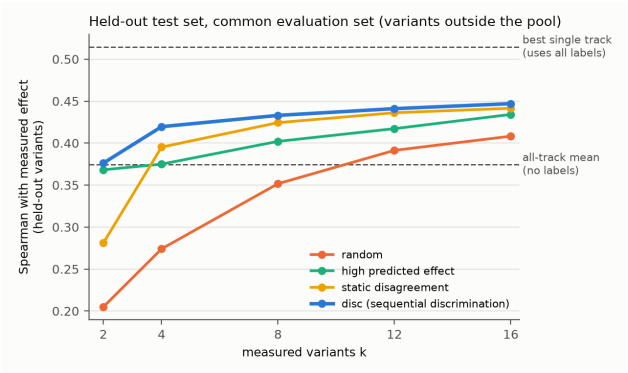


Fig. 2. Kircher test set (C3): Spearman correlation on held-out variants against the number of measured variants k. All panels use the same track posterior. Dashed lines are the label-free all-track mean and the best single track chosen with all labels (a reference).

Table 2. Kircher test set, C1 (all tracks kept), 11 regions. Difference from the label-free matched-track mean (0.405); harmed = regions worse by more than 0.05. v1 and the controls were analysed after the v1 pre-registration; v2 was frozen on the development set before this run.

	$k = 8$		$k = 16$	
Method	diff.	harmed	diff.	harmed
v1 (no anchor)	−0.029	7	−0.014	4
v2 (anchor, frozen)	+0.001	1	+0.018	0
disc panel + `meta_ridge`	+0.002	–	+0.018	–
random panel + `matched_cal`	−0.079	–	−0.043	–

3.2. When matched tracks exist, the prior must default to them

The C3 setting is a stress test. In practice, a laboratory working in a profiled cell type would simply use the matched tracks. When we kept all tracks (C1), the anchor-free method (v1) was on average 0.029 below the label-free matched mean at $k = 8$, and 7/11 regions were worse by more than 0.05 (post hoc; Table 2). It helped only where the matched tracks were poor (*LDLR*: matched mean 0.24, v1 0.51). It lost where they were good (*SORT1*: 0.65 against 0.48).

The metadata anchor removed most of this harm. With $\pi = 0.95$, chosen on the development set by a rule fixed in advance (and re-chosen in 4/5 leave-one-element-out runs), v2 was level with the matched mean at $k = 8$ (+0.001, 1/11 harmed) and ahead at $k = 16$ (+0.018, none harmed). Rescue was slower: at *LDLR*, v2 reached 0.29 at $k = 8$ and 0.41 at $k = 16$. On the same disc panel, v2 and the simpler `meta_ridge` control were tied on Kircher. Re-fitting the matched mean with eight labels (`matched_cal`) was harmful, because the fitted slope can flip sign (*LDLR* fell to 0.14).

3.3. Pre-registered validation on 125 new loci

We then ran the frozen v2 once on the Siraj et al. data (Table 3, Fig. 3). All four primary hypotheses held. In C1, eight measurements did not harm the matched tracks. The mean difference was +0.010 with a one-sided lower bound of +0.005, against a margin of −0.02. Only 3/125 loci were worse by more than 0.05 (upper bound 6.1%, against 15%). At $k = 16$ the difference was +0.020, and one locus was harmed. In C3, eight measurements improved on the all-track mean by 0.048 (0.391 to 0.440; $P = 8 \times 10^{-13}$). In C2, sixteen measurements improved on the related-cell mean by 0.041 (0.412 to 0.453; $P = 1 \times 10^{-19}$). The pre-registered decision was therefore that the safety-net claim is supported.

The secondary analyses, which report effect sizes only, sharpen this picture. The panel matters most when there is no anchor. In C3, disc beat random panels by 0.093 at 122/125 loci, and with random panels the same inference (0.346) fell below the label-free mean (0.391). In C1, the advantage was 0.018. Unlike on Kircher, v2 was slightly ahead of `meta_ridge` on the same panel (+0.008, 97/125, two-sided $P < 0.001$). Re-fitting the matched mean with eight labels lowered accuracy at every one of the 125 loci (mean −0.079). The family prior had no effect (−0.002, $P = 0.11$); we keep it only because it makes the method invariant to duplicated tracks. The C1 gain was larger in HepG2 (+0.017) than in K562 (+0.003, interval including 0).

3.4. Dependence on the model's training exposure

The distilled ALL_FOLDS checkpoint was trained on every genomic region, including those studied here. AlphaGenome also provides four checkpoints, each trained without one fold of the genome. We reran the Kircher analysis with, for each element, the checkpoint that never trained on its region. The specification and reading rule were committed before the runs. All absolute accuracies fell by about 0.035; for example, the best single track fell from 0.465 to 0.428. The advantage over random panels became larger: +0.093, 10/11 regions, $P = 0.0015$. The advantage over the all-track mean shrank to +0.054 (7/11, $P = 0.062$), mainly at *IRF4*, *MSMB* and *LDLR*. The difference between v2 and the matched mean moved from +0.001 to −0.006, with one harmed region in both cases. By our pre-declared rule, which required every contrast to keep its sign, we report these conclusions as dependent on training exposure. In hindsight, a sign rule is too strict for a non-inferiority quantity that is near zero.

Table 3. Pre-registered Siraj test, frozen v2, 125 loci (two cell types averaged per locus). D is the mean paired difference in Spearman correlation.

Hyp.	Comparison	D	Criterion, result
P1	C1, $k = 8$: v2 − matched mean	+0.010	LB $+0.005 > -0.02$, holds
P2	C1, $k = 8$: loci with $D < -0.05$	3/125	UB $0.061 \leq 0.15$, holds
P3	C3, $k = 8$: v2 − all-track mean	+0.048	≥ 0.03, $P = 8\times10^{-13}$, holds
P4	C2, $k = 16$: v2 − related mean	+0.041	> 0, $P = 1\times10^{-19}$, holds
S1	C3, $k = 8$: disc − random panel	+0.093	$122/125 > 0$
S2	C1, $k = 8$: v2 − `meta_ridge`	+0.008	$97/125 > 0$
S3	C1, $k = 8$: `matched_cal` − matched	−0.079	$0/125 > 0$
S5	C1, $k = 8$: family − uniform prior	−0.002	$P = 0.11$

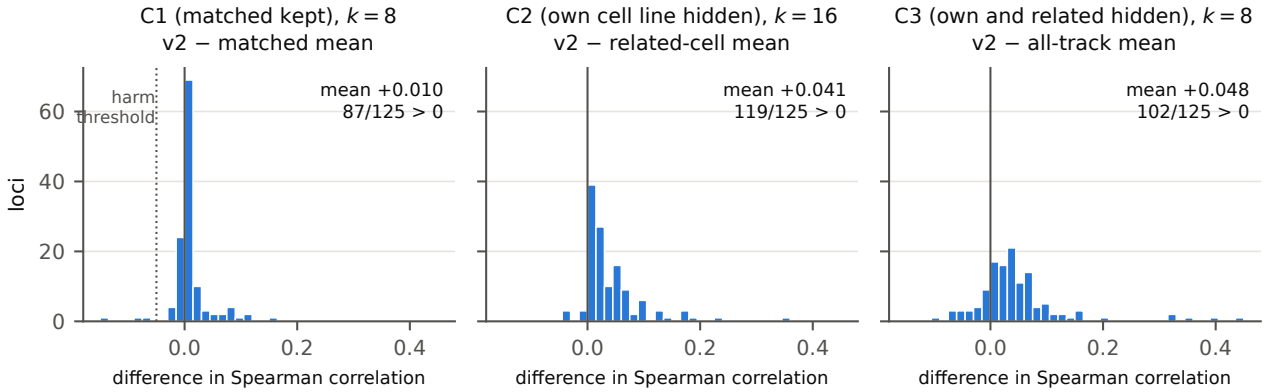


Fig. 3. Per-locus paired differences on the 125 Siraj loci for the three pre-registered primary comparisons. The dotted line marks the pre-registered harm threshold (−0.05).

Eleven regions give little power, so we repeated the check on the 125 Siraj loci. We used the thresholds of the Siraj pre-registration and a reading rule committed before scoring. Each locus overlaps the training intervals of exactly one fold, so each has a unique held-out checkpoint. Absolute accuracies fell again, by 0.04–0.07. The matched-track mean fell from 0.474 to 0.402, v2 with eight labels from 0.483 to 0.426, and the best single track from 0.535 to 0.492. With matched and related tracks hidden, the all-track mean fell from 0.391 to 0.347 and v2 from 0.440 to 0.398. All four pre-registered hypotheses still held, with larger effects (Supplementary Table [TODO: S]). P1: +0.024, one-sided lower bound +0.017, 1/125 loci harmed. P3: +0.052, $P = 3 \times 10^{-13}$. P4: +0.054, $P = 7 \times 10^{-21}$. Paired over loci, the change from ALL_FOLDS was +0.015 for P1 (95% bootstrap interval +0.008 to +0.022), +0.014 for P4 (+0.006 to +0.021) and +0.003 for P3 (−0.006 to +0.013). The matched tracks lost the most accuracy without training exposure, and the measurements recovered part of that loss. On this larger set, the advantage over the all-track mean did not weaken as it did on the 11 Kircher regions.

4. Discussion

Foundation models now give laboratories a thousand candidate predictions for every variant. Our results suggest a simple protocol for choosing among them. Start from the tracks whose metadata match the cell type. Measure about eight variants chosen to split the competing tracks. Let the measurements move the prediction away from the default only when they contradict it. On 125 new loci, this cost nothing on average when the default was available: the mean difference was slightly positive, and 3/125 loci lost more than 0.05. It recovered 0.04–0.05 when the default was missing or only approximate.

Three findings are, in our view, more general than the method. First, *which* variants are measured matters as much as how they are modelled. With random measurements, the same Bayesian inference was worse than not measuring at all. Second, the obvious shortcut of re-fitting the matched track to the new measurements is harmful with eight labels, because the slope is poorly determined and can flip. A prior centred on the metadata default avoids this. Third, the output should be an *average*. With matched tracks kept on Kircher at $k = 8$, the posterior-weighted prediction was ahead of the single most probable track by 0.045 for v1 and 0.007 for v2. At *HBG1* it even exceeded the best single track chosen with all labels. Reporting one proxy track therefore discards information.

The method is closely related to label-efficient model selection for pretrained classifiers [Karimi et al., 2021, Okanovic et al., 2025, Kay et al., 2025, Durmazkeser et al., 2025]. It differs in three ways: candidates are regression outputs with an unknown per-candidate calibration, which we integrate out; the output is a model average rather than a single selected model; and the candidates are about a thousand strongly correlated heads of one network. Our acquisition rule is the classical discrimination criterion [Box and Hill, 1967]. Its value relative to simpler targeted panels is modest: H3 was not significant, and static disagreement was close behind on Kircher. Batch designs that choose all variants at once, and information-based criteria [Rainforth et al., 2024], are natural next steps. [TODO: batch panel and stopping-rule results, if added]

Limitations. K562 and HepG2 are profiled by AlphaGenome. For Siraj, the situations without matched tracks are simulated by hiding tracks, not observed in truly unprofiled cells. The evaluation

is retrospective, and a prospective panel may behave differently. Absolute correlations are moderate (0.44–0.49 on Siraj), and gains are small where the assay has few significant variants. On Kircher, v2's anchor did not beat the simpler metadata-centred ridge; on Siraj it was only slightly ahead. The stopping rule stalls under the anchor prior and needs redesign. The checkpoint analysis shows that training exposure inflates absolute accuracy by 0.04–0.07. On the 11 Kircher regions the advantage over the label-free mean weakened on held-out folds; on the 125 Siraj loci it did not. Our held-out folds are defined by the element's own window; the model's 1 Mb context still reaches sequence from other folds.

5. Conclusion

A handful of well-chosen measurements is enough to decide, among about a thousand foundation-model tracks, which to trust for a cell type. This holds as long as the decision starts from the metadata default and the measurements are chosen to discriminate. In two pre-registered tests, the approach was safe when matched tracks exist and improved predictions when they do not, with eight to sixteen measured variants.

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[TODO: acknowledgements] [TODO: Required by the journal's LLM policy, to be completed by the authors: declare where AI tools were used (for example code, analysis scripts, figures, language editing), in this section or in Methods and in the cover letter. The journal does not accept papers drafted with LLMs; the final text must be written by the authors.]

Author contributions

[TODO: CRediT roles for each author]

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Conflict of interest

None declared.

Data availability

Kircher et al. data are available from the original publication [Kircher et al., 2019] [TODO: database accession]. Siraj et al. saturation mutagenesis data are available from Zenodo under CC-BY-NC [Siraj et al., 2025]; they are not redistributed. AlphaGenome predictions can be regenerated with the public API and the scripts in the repository. Pre-registrations, code and all derived tables are at `https://github.com/ydvislab/AuraGenome-V`.

References

Žiga Avsec, Vikram Agarwal, Daniel Visentin, et al. Effective gene expression prediction from sequence by integrating long-range interactions. *Nature Methods*, 18:1196–1203, 2021. doi: 10.1038/s41592-021-01252-x.

Žiga Avsec, Natasha Latysheva, Jun Cheng, Guido Novati, et al. Advancing regulatory variant effect prediction with AlphaGenome. *Nature*, 649:1206–1218, 2026. doi: 10.1038/s41586-025-10014-0.

G. E. P. Box and W. J. Hill. Discrimination among mechanistic models. *Technometrics*, 9:57–71, 1967. doi: 10.1080/00401706.1967.10490441.

Yavuz Durmazkeser, Patrik Okanovic, Andreas Kirsch, Torsten Hoefler, and Nezihe Merve Gürel. Large language model selection with limited annotations. arXiv:2510.09418, 2025.

ENCODE Project Consortium, Federico Abascal, Reyes Acosta, et al. Expanded encyclopaedias of DNA elements in the human and mouse genomes. *Nature*, 583:699–710, 2020. doi: 10.1038/s41586-020-2493-4.

Jennifer A. Hoeting, David Madigan, Adrian E. Raftery, and Chris T. Volinsky. Bayesian model averaging: a tutorial. *Statistical Science*, 14:382–417, 1999. doi: 10.1214/ss/1009212519.

Connie Huang, Richard W. Shuai, Parth Baokar, et al. Personal transcriptome variation is poorly explained by current genomic deep learning models. *Nature Genetics*, 55:2056–2059, 2023. doi: 10.1038/s41588-023-01574-w.

Mohammad Reza Karimi, Nezihe Merve Gürel, Bojan Karlaš, Johannes Rausch, Ce Zhang, and Andreas Krause. Online active model selection for pre-trained classifiers. In *Proceedings of the 24th International Conference on Artificial Intelligence and Statistics (AISTATS)*, 2021. arXiv:2010.09818.

Justin Kay, Grant Van Horn, Subhransu Maji, Daniel Sheldon, and Sara Beery. Consensus-driven active model selection. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, 2025. arXiv:2507.23771.

Martin Kircher, Chenling Xiong, Beth Martin, et al. Saturation mutagenesis of twenty disease-associated regulatory elements at single base-pair resolution. *Nature Communications*, 10:3583, 2019. doi: 10.1038/s41467-019-11526-w.

Johannes Linder, Divyanshi Srivastava, Han Yuan, Vikram Agarwal, and David R. Kelley. Predicting RNA-seq coverage from DNA sequence as a unifying model of gene regulation. *Nature Genetics*, 57:949–961, 2025. doi: 10.1038/s41588-024-02053-6.

Patrik Okanovic, Andreas Kirsch, Jannes Kasper, Torsten Hoefler, Andreas Krause, and Nezihe Merve Gürel. All models are wrong, some are useful: Model selection with limited labels. In *Proceedings of the 28th International Conference on Artificial Intelligence and Statistics (AISTATS)*, 2025. arXiv:2410.13609.

Tom Rainforth, Adam Foster, Desi R. Ivanova, and Freddie Bickford Smith. Modern Bayesian experimental design. *Statistical Science*, 39:100–114, 2024. doi: 10.1214/23-STS915.

Alexander Sasse, Bernard Ng, Anna E. Spiro, et al. Benchmarking of deep neural networks for predicting personal gene expression from DNA sequence highlights shortcomings. *Nature Genetics*, 55:2060–2064, 2023. doi: 10.1038/s41588-023-01524-6.

Dustin Shigaki, Orit Adato, Aashish N. Adhikari, Shengcheng Dong, et al. Integration of multiple epigenomic marks improves prediction of variant impact in saturation mutagenesis reporter assay. *Human Mutation*, 40:1280–1291, 2019. doi: 10.1002/humu.23797.

Layla Siraj, Rodrigo Ivan Castro Lopez Vaal, Hannah Dewey, Susan Kales, et al. Functional dissection of complex and molecular trait variants at single nucleotide resolution [dataset]. Zenodo, 2025.

Layla Siraj, Rodrigo I. Castro, Hannah B. Dewey, Susan Kales, et al. Functional dissection of complex trait variants at single-nucleotide resolution. *Nature*, 2026. doi: 10.1038/s41586-026-10121-6.

The FANTOM Consortium and the RIKEN PMI and CLST (DGT). A promoter-level mammalian expression atlas. *Nature*, 507:462–470, 2014. doi: 10.1038/nature13182.

Robert Tibshirani. Regression shrinkage and selection via the lasso. *Journal of the Royal Statistical Society Series B: Statistical Methodology*, 58:267–288, 1996. doi: 10.1111/j.2517-6161.1996.tb02080.x.

Jian Zhou and Olga G. Troyanskaya. Predicting effects of noncoding variants with deep learning-based sequence model. *Nature Methods*, 12:931–934, 2015. doi: 10.1038/nmeth.3547.